

Type of Irrigation

surface irrigation
subsurface irrigation

- ① Surface irrigation: water applied/distributed over soil surface by either gravity or by pumping
- Flow irrigation → water flow by gravity from high to low level
 - lift irrigation → water lift up by mechanical equipment (like pump) then supply for irrigation.

Type of Flow irrigation:-

- ① Perennial Irrigation
- constant and continuous water-supply throughout the crop period
 - Direct irrigation → direct divert river water to main canal by constructing weir/barrage.
 - Storage irrigation → construct dam
- ② Flood Irrigation or uncontrolled irr. or Inundation irr.
- soil kept submerged & thoroughly flood with water so to cause thorough saturation of land.

② Subsurface irrigation: water flows underground and nourish plant roots by capillarity.

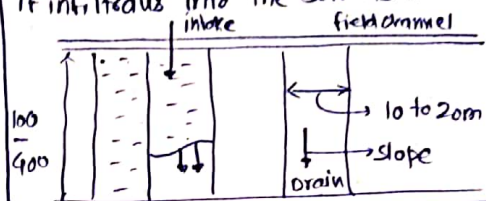
- natural subsurface irrigation
- Artificial subsurface irrigation

Method of Irrigation:-

- ① Free Flooding (wild flooding)
- 
- After water leaves the ditches, no attempt is made to control the flow by means of Levees → ∴ wild flooding
 - suit for close growing crops, pasture particularly where land is steep
 - method may be used on rolling land (topography irregular { where border's, check, furrows are not suitable })

② Border method

- Land divided into no. of strips & separated by low levees called ~~strip~~ Borders.
- Strip Length = 100 to 400m width 10-20m.
- water is allowed to flow from the supply - ditch into each strip. The water flows slowly towards the lower end and it infiltrates into the soil as it advances.



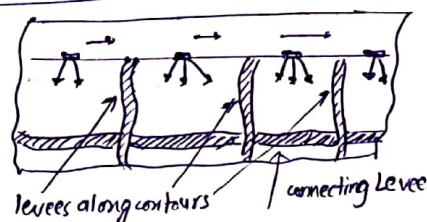
$$t = \frac{y}{i} \ln \frac{Q}{Q - iA}$$

depth over border strip
area of land strip
Infiltration rate

$$A_{max} = \frac{Q}{i}$$

Surface flow will stop after irrigation of this area & deep percolation will start.

③ check flooding



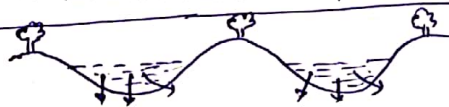
- similar to ordinary flooding except that the water is controlled by surrounding the check area with low and flat levees.

Best for → close growing crops (jowar, paddy)

→ deep homogeneous loam or clay soil with medium infiltration rate are preferred

→ suit for both less & more permeable soil

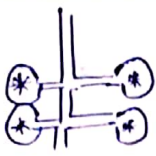
④ furrow method



crops are usually grown on the ridges b/w the furrows

- Best for row crops { maize, cotton, potatoes, sugarcane, ground nut, tobacco }

⑤ Basin Flooding



- Best for Orchard Tree
- coarse sand not suitable because of more percolation loss

⑥ Drip Irrigation Method or Trickle Irrigation

- Popular in areas where scarcity of irrigation water and salt problem.
- in this water & fertilizer is slowly and directly applied to the root zone of plants, to minimize the loss due to evaporation & percolation

Best for row crops and orchards such as tomatoes, grapes, corn, citrus, melon etc.

⑦ Sprinkler Irrigation method

- In this, irrigation water is applied to the land in form of spray. ~~under network of pipe &~~ ^{30 high water demand}
- used for all crops except rice & jute.
- suit for all soils except for very heavy soil (soil with low infiltration rate)
- Best suited for very light soil as deep percolation losses are avoided { light soil → these soil drain quickly after rain & watering }
- this method mainly by cultivators of Tea, coffee and vegetables.
- This system is flexible to suit undulating Topography (hence levelling is not necessary)

Best conditions for This method

- ① Topography irregular (undulating)
- ② land gradient steeper & soil easily erodable
- ③ soil excessive permeable
- ④ water table high
- ⑤ when water availability is difficult & scarce

Irrigation water Quality :-

① Proportion of sodium ion into cations
↓
ESP SAR

(i) Exchangeable sodium percentage (ESP)

$$ESP = \frac{Na^+}{Na^+ + Mg^{+2} + K^+ + Ca^{+2}}$$

(ii) sodium adsorption ratio (SAR)

$$SAR = \frac{Na^+}{\sqrt{\frac{Ca^{+2} + Mg^{+2}}{2}}}$$

→ ecm (equivalent per million)

मिली इकाई ⇒ $\frac{mg/l \times 1}{(\text{atomic wt} / \text{valency})}$

category	SAR	Type of water	
S ₁	0-10	Low sodium water	for all soil crops
S ₂	10-18	medium "	coarse grains soil & organic soil with good permeability
S ₃	18-26	High "	Drainage + Leaching (gypsum)
S ₄	> 26	very high	not suitable

② total conc. of soluble salt

→ electrical conductivity (EC)

EC (mho/cm) @ 25°C	class	Salinity	Remarks
0-250	C ₁	Low	for all crops
250-750	C ₂	medium	all crop if leaching done
750-2250	C ₃	High	for High salt tolerant plant with special measure to control salinity
> 2250	C ₄	very high	not for irrigation

③ Sediment concentration

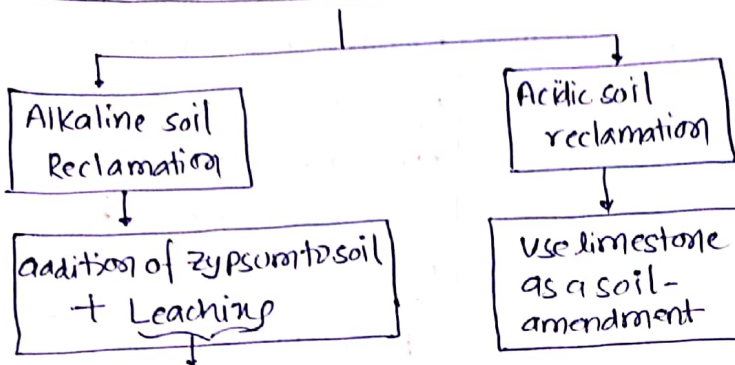
④ conc. of potential toxic element (Boron, selenium)

Boron > 0.3 ppm. if

⑤ Bicarbonate conc.

⑥ Bacterial contamination

Reclamation → Process by which uncultivated land made fit for cultivation.



Leaching: in this land flooded with adequate depth of water, the Alkali-salts present in the soil, get dissolved in this water which percolate down to join the water table or drained away by surface and subsurface drains, This process known as Leaching.

• This process is repeated till the salts in the top layer of the land are reduced to such extent that some salt-resistant crop can be grown.

Saline soil → land affected by efflorescence

Alkali salt → injurious

- least harmful → NaCl
- most " → Na_2CO_3

Na_2CO_3
 NaCl
 Na_2SO_4

note: - when Na_2CO_3 is present (मृदा अम्लीय soil)
 ∴ add gypsum before leaching

High salt resistant crop → fodder
 Berseem
 Bajra

Alkaline soil ∴ if efflorescence continue for long time → base reaction happens.

if soil clay then due to base rxn → sodiumising clay making it impermeable, illaerated, highly unproductive

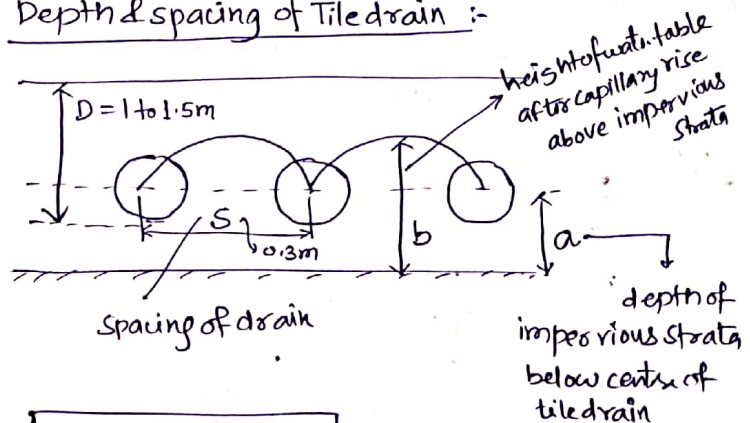
Land capability → US beaurso of reclamation classification

Leaching requirement = $LR = \frac{D_i - C_u}{D_i}$ → Consumption Use + drain water out

$LR = \frac{D_d}{D_i} = \frac{D_i - C_u}{D_i} = \frac{(EC)_d}{(EC)_i} = \frac{C_i}{C_d}$ total irrigation water applied.

DC (drainage coefficient) ∴ एक ही में पानी की height.
 → used for calculation of volume of water $\left(\frac{\text{m}^3}{\text{sec}}\right)$

Depth & spacing of Tile drain ∴



$S = \frac{4k}{q} (b^2 - a^2)$

$P = 1\%$ of Annual avg. rainfall

$q = \left(\frac{P/100}{24 \times 3600} \right) [S \times I] \rightarrow \text{length of tile drain}$

or $q = \left\{ \text{drainage coeff.} \times \text{spacing} \times \text{Lgh of tile drain} \right\}$
 $\left\{ 0.116 \frac{\text{m}^3}{\text{km}^2} \right\} \times (12 \times 11) \times 1$

Duty :- area of land that can be irrigated by unit volume of irrigation water

(D) 'or' area of land in hectares that can be irrigated if 1 cumec of water is supplied continuously to the land throughout the Base period.

$$D = \frac{8.64 B(\text{days})}{\Delta(\text{metr})}$$

↓
hectare
cumec

note:
in-tank irrigation
Duty = $\frac{h}{q} \text{ m}^3$

Base period :- time b/w first watering to last watering.
(B)

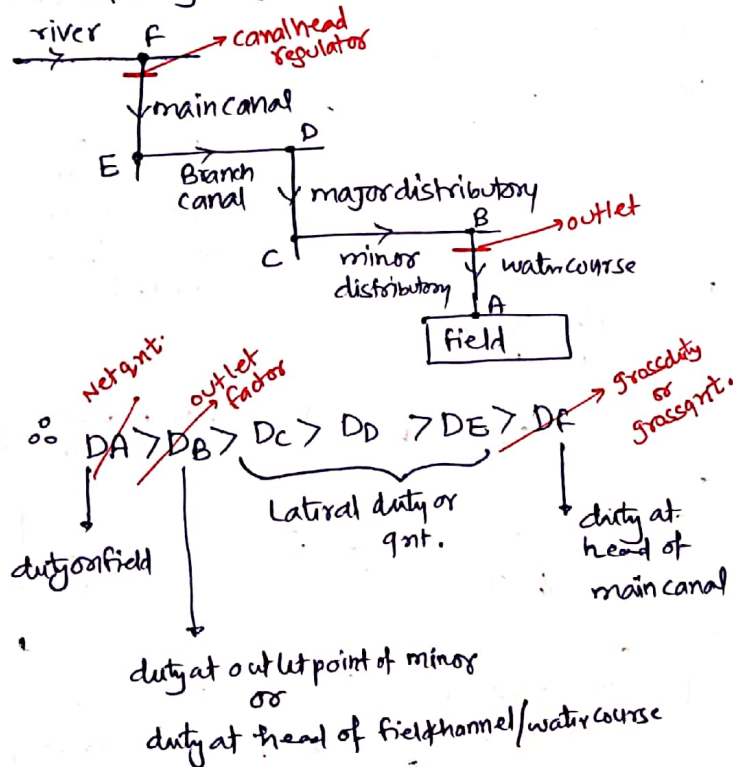
↓
to prepare land
for sowing crop

↓
before its
harvesting

note: crop period $\Rightarrow$ sowing to harvesting time.

Delta (Δ) :- total depth of water over the irrigated land req. by crop during its entire Base period.

note: as one moves from head of canal to field $\rightarrow$ duty increases (मतलब downstream जाने पर duty बढ़ती जाती है)



Factors affecting duty :-

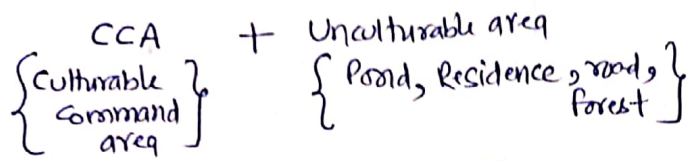
① Type of crop	if qnt of water req. $\uparrow \therefore D \downarrow$ (loss due to ET $\uparrow$)
② climate condition	* Temperature : $T \propto \frac{1}{D}$ * Velocity (wind) $V \propto \frac{1}{D}$ * Humidity $H \propto D$ (loss due to ET $\downarrow$) * Rainfall : $P \propto D$
③ system of irrigation	Perennial D $>$ Foundation D
④ method of irrigation	furrow D $>$ any other D surface method D $<$ Sprinklers, drip D
⑤ Quality of irrigation water	Salt $\uparrow \Rightarrow D \downarrow$ * अधिकतर पानी leaching में बना जायेगा
⑥ method of cultivation	Proper ploughing $\therefore D \uparrow$
⑦ Time of irrigation and frequency of irrigation	at initial stage more water required hence D $\downarrow$ later D $\uparrow$
⑧ Type of soil & subsoil of area through which canal passes	
⑨ Base period	B $\uparrow \Rightarrow$ more water req. hence D $\downarrow$
⑩ canal conditions	earthen canal $\Rightarrow D \downarrow$ lined canal $\Rightarrow D \uparrow$

Season	time	फसल
रबी	oct - march	गेहूँ, मटर, चना, सरसो, आलू
खरीफ	Apr - sept	ज्वार, झुंगफली, कपास, जूट
Zaid period	march - June	
Perennial crop	साल भर	Sugarcane.

Important Terms :-

① Command area → By a canal system how much area is irrigated.

② Gross Command area (GCA) :



③ CCA (Culturable command area) = GCA - Unculturable

Culturable
Cultivated area

Position of CCA
which is actually cultivated
during crop season.

Culturable
uncultivated
area

→ not cultivated

④ Intensity of irrigation : % of CCA proposed to be irrigated annually.

Ex. if intensity of Irr. for Rabi = 50%
Kharif = 65%

∴ yearly intensity of irr = 50 + 65 = 115%

⑤ Crop ratio : (Rabi-Kharif ratio) ⇒ Ratio of land irrigated during 2 main crop seasons Rabi & Kharif

$$CR = \frac{\text{CCA under Rabi}}{\text{CCA under Kharif}}$$

⑥ Paleo irrigation : watering done prior to sowing of crop.

↳ This is done to prepare the land for sowing and to add sufficient moisture to the soil which would be required for the initial growth of the crop.

⑦ Time factor (T.F) = $\frac{\text{no. of days canal run}}{\text{Irrigation period}}$ $\frac{140}{180}$

$$Q_{\text{design}} = \frac{Q}{T.F}$$

⑧ Capacity factor (CF) = $\frac{\text{MSR of canal}}{\text{FSR}}$

Ex. in kharif if discharge required = 0.95 Q_{max}
∴ CF = 0.95

⑨ Cash crops : encashed in marketing for processing.

{ Can not directly consumed by cultivator }

Ex. { गन्ना, चाय, ~~काका~~, कॉफी, तम्बाकू }

⑩ Crop calendar : tool provides information about planting, sowing, harvesting of locally-adopted crop in an area.

concept : KOR :-

① Kor watering :- 1st watering after plants have grown a few centimeters high.

② Kor Depth :- Depth of water Applied during This watering (Kor watering)

③ Kor period : Kor watering must be done in a limited period which is known as Kor Period

Perimeter :-	Crop	Day	Kor depth (cm)
		Kor period	
	Rice	14 day	19 cm
Intl.	wheat	28 day	14 cm

Irrigation efficiency:

① water-conveyance efficiency

$$\eta_c = \frac{\text{Field water}}{\text{diverted from river into canal}}$$

• It accounts for the water losses which occur in conveyance from the point of diversion into the canal system to the field

② Application efficiency

$$\eta_a = \frac{\text{water stored in root zone}}{\text{field water}}$$

• It accounts for the water losses which occur during the application of irrigation water (Ex. surface runoff & deep percolation)

③ water use efficiency

$$\eta_u = \frac{\text{water used beneficially including the water req. for leaching}}{\text{field water}}$$

④ storage efficiency

$$\eta_s = \frac{\text{amt of water stored in root zone}}{\text{water needed to bring the m.c of soil to field capacity}}$$

⑤ water distribution efficiency

$$\eta_d = \left[1 - \frac{y}{d} \right] \times 100$$

$d \rightarrow$ avg. depth of water $\Rightarrow \frac{d_1 + d_2 + \dots + d_n}{n}$

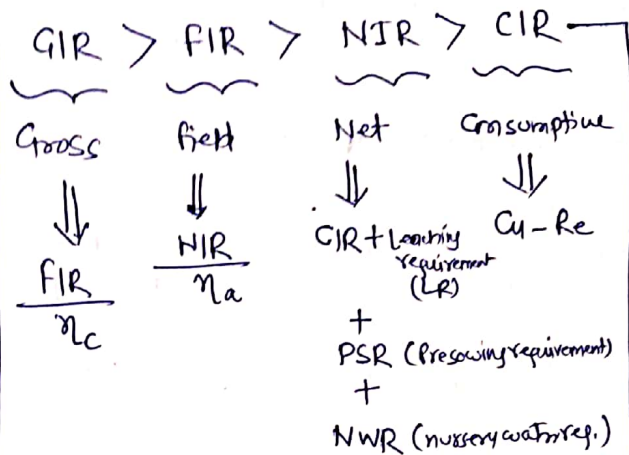
$y \rightarrow$ Avg. numerical Deviation in depth of water

$$\frac{|d_1 - d| + |d_2 - d| + \dots + |d_n - d|}{n}$$

⑥ Consumptive efficiency

$$\eta_{ce} = \frac{\text{water use consumptively}}{\text{net amt of water depleted from root zone}}$$

Irrigation Requirement of crop :-



QR :- amt of irrigation water that is req. to meet the evapotranspiration need of a crop during its full growth.